A_Bhadouria_HW11

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Question 1

Rate of absorption of gases A and B based on surface renewal concept

Gases A, B

$y=0$

Liq $A \rightleftharpoons \alpha M \quad (K_A)$

$B \rightleftharpoons \beta M \quad (K_B)$

y

$y=\infty$

$$K_A = \frac{k_1}{k_{-1}} = \frac{C_M^{\alpha}}{C_A}$$

$$K_B = \frac{C_M^{\beta}}{C_B}$$

(i) A, B, M have the same diffusion coefficients

Mass balance: $\frac{\partial c}{\partial t} = -\nabla \cdot N + \sum \alpha_i r_i$; $r_A = -k_1 C_A + k_{-1} C_M^{\alpha}$; $r_M = k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta}$

$r_B = -k_2 C_B + k_{-2} C_M^{\beta}$

$$\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_A}{\partial y^2} + \frac{r_A}{\alpha} = D \frac{\partial^2 C_A}{\partial y^2} + \frac{(-k_1 C_A + k_{-1} C_M^{\alpha})}{\alpha}$$

Similarly,

$$\frac{\partial C_B}{\partial t} = D \frac{\partial^2 C_B}{\partial y^2} + \frac{(k_2 C_B + k_{-2} C_M^{\beta})}{\beta}$$

$$\frac{\partial C_M}{\partial t} = D \frac{\partial^2 C_M}{\partial y^2} + (k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta})$$

$$\therefore \quad \alpha \frac{\partial C_A}{\partial t} = \alpha D \frac{\partial^2 C_A}{\partial y^2} - k_1 C_A + k_{-1} C_M^{\alpha}$$

$$\beta \frac{\partial C_B}{\partial t} = \beta D \frac{\partial^2 C_B}{\partial y^2} - k_2 C_B + k_{-2} C_M^{\beta}$$

$$+ \quad \frac{\partial C_M}{\partial t} = D \frac{\partial^2 C_M}{\partial y^2} + k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta}$$

$$\frac{\partial}{\partial t}(\alpha C_A + \beta C_B + C_M) = D \frac{\partial^2}{\partial y^2}(\alpha C_A + \beta C_B + C_M)$$

$$\boxed{\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial y^2}}$$

Boundary conditions:

@ $y=0$, $c(0,t) = \alpha C_{A_i} + \beta C_{B_i} + C_{M_i} = C_i$ (Assuming that interfacial conc. is constant throughout time)

@ $t=0$, $c(y,0) = \alpha C_{A_0} + \beta C_{B_0} + C_{M_0}$

@ $y \to \infty$, $c(y \to \infty, t) = \alpha C_{A_0} + \beta C_{B_0} + C_{M_0} = C_0$

(ii) First we solve for the concentration profile:

General solution of PDE's : $c(y,t) = A + B \operatorname{erf}\left(\frac{y}{2\sqrt{Dt}}\right)$

↳ Solving for constants,

$$c(0,t) = c_i = A + B\,\cancel{erf(0)}^{\,0} \Rightarrow A = c_i$$

$$c(\infty,t) = c_0 = A + B\,\cancel{erf(\infty)}^{\,1} = A + B \;;\; B = c_0 - c_i$$

$$\therefore \boxed{c(y,t) = c_i + (c_0 - c_i)\, erf\left(\frac{y}{2\sqrt{Dt}}\right)} \Rightarrow \quad c_A = \frac{c_M^a}{K} \;;\; c_M = c - \alpha c_A - c$$

$$N_A = D\left.\frac{\partial c_A}{\partial y}\right|_{y=0} \;;\; c_A = \frac{c - \beta c_B - c_M}{\alpha} \;;\; N_A = \left.\frac{D\partial c}{\partial y}\right|_{y=0} - \frac{D\beta}{\alpha}\left.\frac{\partial c_B}{\partial y}\right|_{y=0} - \frac{D}{\alpha}\left.\frac{\partial c_M}{\partial y}\right|_{y=0}$$

$$N_B = D\left.\frac{\partial c_B}{\partial y}\right|_{y=0} \;;\; c_B = \frac{c - \alpha c_A - c_M}{\beta} \qquad N_A = D\left.\frac{\partial c}{\partial y}\right|_{y=0} - \frac{\beta}{\alpha}N_B - \frac{1}{\alpha}N_M$$

$$N_M = D\left.\frac{\partial c_M}{\partial y}\right|_{y=0} \;;\; c_M = c - \alpha c_A - \beta c_B$$

$$\therefore \boxed{N_A + \frac{\beta}{\alpha}N_B + \frac{1}{\alpha}N_M = D\left.\frac{\partial c}{\partial y}\right|_{y=0} = N}$$

$$N = D\left.\frac{\partial c}{\partial y}\right|_{y=0} \;;\; \frac{\partial c}{\partial y} = (c_0 - c_i)\frac{2e^{-y^2}}{\sqrt{\pi t}}$$

$$\left.\frac{\partial c}{\partial y}\right|_{y=0} = (c_0 - c_i)\frac{2}{\sqrt{\pi t}}$$

$$\bar{N} = S \cdot H \int_0^\infty \frac{e^{-st}}{\sqrt{t}}\, dt$$

$$= S \cdot (c_0 - c_i)\frac{2}{\sqrt{\pi}} \cdot D \int_0^\infty \frac{e^{-st}}{\sqrt{t}}\, dt$$

$$\boxed{\bar{N} = \sqrt{DS}\,(c_0 - c_i)}$$